File - /Users/mazenfraihat/IdeaProjects/HW10/src/test/java/part1/UserTest.java

```java
package part1;

import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;

import java.util.List;

import static org.junit.jupiter.api.Assertions.*;

class UserTest {

    ChatServer server = new ChatServer();
    User mazen = new User("Mazen", server);
    User ahmed = new User("Ahmed", server);

    @BeforeEach
    void setUp() {
        server.registerUser(mazen);
        server.registerUser(ahmed);
    }

    @Test
    void getName() {
        assertEquals("Mazen", mazen.getName());
    }

    @Test
    void getChatHistory() {
        assertNotNull(mazen.getChatHistory());
    }

    @Test
    void sendMessage() {
        mazen.sendMessage("hi", List.of(ahmed));
        assertEquals(1, mazen.getChatHistory().getMessages().size());
        assertEquals(1, ahmed.getChatHistory().getMessages().size());
    }

    @Test
    void receiveMessage() {
        Message message = new Message(mazen, List.of(ahmed), "hi");
        ahmed.receiveMessage(message);
        assertTrue(ahmed.getChatHistory().getMessages().contains(
message));
    }

    @Test
    void undoLastMessage() {
        mazen.sendMessage("hi", List.of(ahmed));
        mazen.undoLastMessage();
        assertTrue(mazen.getChatHistory().getMessages().isEmpty());
        assertTrue(ahmed.getChatHistory().getMessages().isEmpty());
    }
```

```
53
54      @Test
55      void blockUser() {
56          ahmed.blockUser(mazen);
57          mazen.sendMessage("hi", List.of(ahmed));
58          assertTrue(ahmed.getChatHistory().getMessages().isEmpty());
59      }
60
61      @Test
62      void viewChatHistory() {
63          mazen.sendMessage("hi", List.of(ahmed));
64          assertDoesNotThrow(() -> mazen.viewChatHistory());
65      }
66  }
```

```java
package part1;

import org.junit.jupiter.api.Test;

import java.util.List;

import static org.junit.jupiter.api.Assertions.*;

class MessageTest {

    ChatServer server = new ChatServer();
    User mazen = new User("Mazen", server);
    User ahmed = new User("Ahmed", server);
    Message message = new Message(mazen, List.of(ahmed), "hi");

    @Test
    void getSender() {
        assertEquals(mazen, message.getSender());
    }

    @Test
    void getRecipients() {
        assertTrue(message.getRecipients().contains(ahmed));
    }

    @Test
    void getTimestamp() {
        assertNotNull(message.getTimestamp());
    }

    @Test
    void getContent() {
        assertEquals("hi", message.getContent());
    }

    @Test
    void testToString() {
        String text = message.toString();
        assertTrue(text.contains("Mazen"));
        assertTrue(text.contains("Ahmed"));
        assertTrue(text.contains("hi"));
    }
}
```

File - /Users/mazenfraihat/IdeaProjects/HW10/src/test/java/part1/ChatServerTest.java

```java
package part1;

import org.junit.jupiter.api.Test;

import java.util.List;

import static org.junit.jupiter.api.Assertions.*;

class ChatServerTest {

    ChatServer server = new ChatServer();
    User mazen = new User("Mazen", server);
    User ahmed = new User("Ahmed", server);
    Message message = new Message(mazen, List.of(ahmed), "hi");

    @Test
    void registerUser() {
        server.registerUser(mazen);
        server.registerUser(ahmed);
        server.sendMessage(message);
        assertEquals(1, ahmed.getChatHistory().getMessages().size());
    }

    @Test
    void unregisterUser() {
        server.registerUser(mazen);
        server.registerUser(ahmed);
        server.unregisterUser(ahmed);
        server.sendMessage(message);
        assertTrue(ahmed.getChatHistory().getMessages().isEmpty());
    }

    @Test
    void blockUser() {
        server.registerUser(mazen);
        server.registerUser(ahmed);
        server.blockUser(ahmed, mazen);
        assertTrue(server.isBlocked(ahmed, mazen));
    }

    @Test
    void isBlocked() {
        server.registerUser(mazen);
        server.registerUser(ahmed);
        assertFalse(server.isBlocked(ahmed, mazen));
    }

    @Test
    void sendMessage() {
        server.registerUser(mazen);
        server.registerUser(ahmed);
        server.sendMessage(message);
        assertTrue(ahmed.getChatHistory().getMessages().contains(
```

```
53 message));
54     }
55
56     @Test
57     void sendMessageToBlockedUser() {
58         server.registerUser(mazen);
59         server.registerUser(ahmed);
60         server.blockUser(ahmed, mazen);
61         server.sendMessage(message);
62         assertTrue(ahmed.getChatHistory().getMessages().isEmpty());
63     }
64
65     @Test
66     void undoMessage() {
67         server.registerUser(mazen);
68         server.registerUser(ahmed);
69         server.sendMessage(message);
70         server.undoMessage(mazen, new MessageMemento("hi", message.
   getTimestamp()));
71         assertTrue(ahmed.getChatHistory().getMessages().isEmpty());
72     }
73 }
```

File - /Users/mazenfraihat/IdeaProjects/HW10/src/test/java/part1/ChatHistoryTest.java

```java
package part1;

import org.junit.jupiter.api.Test;

import java.util.List;

import static org.junit.jupiter.api.Assertions.*;

class ChatHistoryTest {

    ChatServer server = new ChatServer();
    User mazen = new User("Mazen", server);
    User ahmed = new User("Ahmed", server);
    Message message = new Message(mazen, List.of(ahmed), "hi");
    MessageMemento memento = new MessageMemento("hi", message.
getTimestamp());
    ChatHistory history = new ChatHistory();

    @Test
    void addMessage() {
        history.addMessage(message);
        assertEquals(1, history.getMessages().size());
    }

    @Test
    void saveMemento() {
        history.saveMemento(memento);
        assertEquals(memento, history.getLastMessageSent());
    }

    @Test
    void getLastMessageSent() {
        assertNull(history.getLastMessageSent());
    }

    @Test
    void removeMessage() {
        history.addMessage(message);
        history.removeMessage(memento);
        assertTrue(history.getMessages().isEmpty());
    }

    @Test
    void getMessages() {
        history.addMessage(message);
        assertTrue(history.getMessages().contains(message));
    }
}
```

```java
package part1;

import org.junit.jupiter.api.Test;

import java.time.LocalDateTime;

import static org.junit.jupiter.api.Assertions.*;

class MessageMementoTest {

    LocalDateTime time = LocalDateTime.now();
    MessageMemento memento = new MessageMemento("hi", time);

    @Test
    void getContent() {
        assertEquals("hi", memento.getContent());
    }

    @Test
    void getTimestamp() {
        assertEquals(time, memento.getTimestamp());
    }
}
```